SIMATS ENGINEERING

ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1519

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1 AT3 Short Answer Test.

Name: Jannathul Firdaus.N

Reg.NO: 192573025

Course: CSA1519 Cloud computing and Big Data Analytics

Date: 13/10/26

Questions:

1. Define cloud computing and Explain four characteristics.
2. Trace the evolution of cloud computing from hardware to software evolution
3. Explain role of server virtualization and difference between cloud computing.
4. Discuss role of data centres.
5. Differentiate Iaas, Paas, Saas, Xaas.

Answers:

cloud computing:

It is defined as the delivery of computing resources such as server, software and networking from devices over the Internet on demand.

characteristics of cloud computing:

On demand self service :

Users can able to utilize the resources whenever required.

Broad network access :

Users can able to access the Services accessed through the internet from different devices.

Resource pooling :

The users can able to share the computing devices among multiple users.

Rapid elasticity :

Resources can be increased or decreased quickly according to the demand.

2. Evolution of cloud computing :

Hardware evolution :

Faster processor, improved Storage high memory and high computational power in hardware evolution.

Server Evolution:

Powerful servers can able to share the computing resources and applications among multiple users.

Distributed systems:

Multiple connected computers worked together to improve the performance and reliability of the system.

Software Evolution:

Faster and wider connectivity enables remote access to the computing resources.

Internet software evolution:

Web application APIs, web server and other technologies are together accessed for the easier and fast access of the computing resources.

3 Server virtualization:

Server virtualization divides one physical server into into multiple virtual machine. Each VM perform their own application and ~~Server~~. Operating system.

* Improve hardware utilization
* Allows multiple users to share the physical resources.
* Easy flexible resource allocation
* making server easier to create and manage.

Virtualization	cloud computing
divides one physical server into multiple Virtual Machines	delivers computing resources over the internet on demand.
focus on efficient use of physical hardware	focus on sharing resources and applications to users.
Ex: multiple VM on one physical server	renting a physical server from AWS.

4. **Data centre:**

A facility containing Servers, storage system, networking equipment, power system and cooling system.

Roles of Data centres:

Resource hosting:

Data centres contain the physical server and storage used to provide cloud services.

Data storage:

They store applications, databases, files and other user information.

Networking:

High speed network connections allow users to access cloud services from different locations.

Reliability & availability:

Backup systems, multiple server and redundant equipment help keep services available even when some components fail.

Security:

Data centres use physical security, monitoring, access control to protect cloud infrastructure and data.

5.

Service model	Uses	Example.
Infrastructure as a service (IaaS)	provides Virtual Machines, storage and networking to user.	Amazon EC2.
Platform as a service (PaaS)	Provides a development platform runtime, tools and infrastructure	Google App Engine.
Software as a service (SaaS)	provides a ready to use software through the internet	google workspace.
Anything / Everything as a service (XaaS)	provides specialized services delivered through the cloud	Amazon RDS.

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